Huawei Access Node QoS Congestion Event Troubleshooting SOP (BAS-ACC-37)

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4494

Description

This Standard Operating Procedure (SOP) addresses QoS congestion events on Huawei Access Nodes, specifically on BAS-ACC-37 located at Basavanagudi POP. QoS congestion events, characterized by microbursts, interface tail drops, and QoS queue starvation, are primarily caused by oversubscription of bandwidth and misconfigured QoS policies. On Huawei Access Nodes, QoS policies are enforced by the Hierarchical Policy and Charging System (HPCS), which evaluates and prioritizes traffic based on configured rules and thresholds. Insufficient bandwidth allocation or incorrect priority settings can lead to congestion and associated alarm ALARM_4494. This SOP outlines the necessary steps to identify and rectify the root cause of QoS congestion events on BAS-ACC-37.

Pre-requisite Checks

- task: Verify redundant link status, description: Ensure the redundant link is online and functioning correctly to isolate the issue to the Access Node.

- task: Validate environmental factors, description: Check for any environmental factors such as temperature fluctuations, power supply issues, or equipment malfunctions that may be contributing to the QoS congestion event.

- task: Gather historical data, description: Collect historical data on QoS congestion events, including timestamps, alarm codes, and associated configuration changes to identify patterns and potential root causes.

Potential Root Causes

Cause	Probability
Oversubscription of bandwidth	High
Misconfigured QoS policies	Medium
Incorrect priority settings	Medium
Hardware or software issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-queue queue-index 1, Expected Output: Queue Index 1: Current Queue Size: 1000 Threshold: 500 , Description: This command displays the current queue size and threshold for queue index 1, helping to identify if the queue is experiencing congestion.

Command: display policy policy-index 1, Expected Output: Policy Index 1: Priority: High Bandwidth: 10000 , Description: This command
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```
displays the policy index 1, including priority and allocated bandwidth, to verify if the QoS policy is correctly configured.

Command: display interface interface-index 1, Expected Output: Interface Index 1: Status: Up Bandwidth: 10000 , Description: This command displays the status and bandwidth allocation for interface index 1, helping to identify if the interface is functioning correctly.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: display policy policy-index 1, Description: Verify that the QoS policy is correctly configured, including priority and allocated bandwidth, to ensure that traffic is properly prioritized.

Step 2: Step: Adjust bandwidth allocation, Command: set interface interface-index 1 bandwidth 5000, Description: Adjust the bandwidth allocation for interface index 1 to increase available bandwidth and help alleviate congestion.

Step 3: Step: Reconfigure QoS policy, Command: set policy policy-index 1 priority Low, Description: Reconfigure the QoS policy to assign a lower priority to the affected traffic, reducing its share of bandwidth and helping to prevent congestion.

Post-Check Verification

```
Command: display current-queue queue-index 1, Expected Output: Queue Index 1: Current Queue Size: 500 Threshold: 500 , Description: Verify that the queue size and threshold have stabilized and are within acceptable limits.

Command: display policy policy-index 1, Expected Output: Policy Index 1: Priority: Low Bandwidth: 5000 , Description: Verify that the QoS policy has been successfully reconfigured with the correct priority and bandwidth allocation.

Command: display interface interface-index 1, Expected Output: Interface Index 1: Status: Up Bandwidth: 5000 , Description: Verify that the interface is functioning correctly with the adjusted bandwidth allocation.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo set policy policy-index 1 priority Low, Description: Restore the original QoS policy configuration to prevent further disruption to traffic.

- Step: Restore original bandwidth allocation, Command: undo set interface interface-index 1 bandwidth 5000, Description: Restore the original bandwidth allocation to prevent further disruption to traffic.